Muhammad Azzazy

US Citizen

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Education

The American University in Cairo (AUC) — Cairo, Egypt
Bachelor of Science in Computer Engineering (GPA: 3.57/4.0) — *Sep. 2020 – Jan. 2025*

Egypt British International School (EBIS) — Cairo, Egypt
A Levels & IGCSEs — *Sep. 2017 – Jun. 2020*

Experience

Full Stack Engineer — April 2026 – Present
Deloitte Innovation Hub (DIH) — *New Cairo, Cairo, Egypt*

- Collaborated with manufacturing business analysts, full stack engineers, and AI/data engineers in an offsite week to gain consulting skills, presentation skills, and other soft skills
- Engineered an AI-powered ecommerce platform as part of a team of full stack, AI & data, and cloud engineers using Spring Boot, React (TypeScript), and Tailwind and integrated Google Agent Development Kit (ADK) agents written in Python to support product recommendations, order tracking, and clothing try-on
- Provided technical full-stack engineering support for the Analyst Induction Program (AIP) during the technical bootcamp weeks
- Earned the *Most Frugal Idea Award* in the *Deloitte Global MashUp Hackathon* after building LUMOS which combines AI and a Raspberry Pi to help people with visual impairments detect their surroundings at the office
- Earned the *DIH x UNICEF NextStep Mentorship Program Top Performer Award* for program facilitation by automating Slack workspace on-boarding using Python, Slack agents and workflows, and Azure Logic Apps

Operations Engineer — November 2025 – April 2026
Huawei — *New Cairo, Cairo, Egypt*

- Monitored network elements for South Africa (SA) Telkom, identifying and escalating service failures using Huawei HEDS tools
- Tracked and reported active incidents using customer Citrix ticketing systems and Excel, delivering hourly CRM status updates
- Collected MSISDNs of users who subscribed within the past week using awk for text processing
- Collaborated with the SA Telkom frontline and Service Delivery Manager (SDM) in Change Configuration Board (CCB) meetings to present and validate Measurements of Procedure (MOPs) for business-driven network changes, ensuring controlled and compliant implementation
- Performed daily Standard Operating Procedure (SOP) checks on the status of 31 Linux nodes by investigating alarms and graphs shown on Huawei iManager I2000; and checking memory utilization, disk utilization, CPU utilization, system logs, and zombie processes
- Automated the process of querying and preprocessing the data from Oracle 11g database for weekly reports for the Front-Office (FO) Lead using Python and cron

Research Assistant — July 2025 – November 2025
The American University in Cairo — *New Cairo, Cairo, Egypt*

- Engineered an information retrieval tool using Python and n8n for fetching metadata of journal articles based on a list of keywords in a JSON file and publication year
- Led the full-stack development of *FebriLogic* which is a web application based on a novel algorithm that uses real-world data to assist healthcare professionals in the differential diagnosis of acute febrile illnesses (AFIs)
- Disseminated findings from the clinical diagnosis support tool, *FebriLogic*, in a scholarly paper submitted to *PLOS Digital Health*

Full Stack Engineering Intern — February 2025 – July 2025
PricewaterhouseCoopers (PwC) Egypt Technology & Innovation Center (ETIC) — *New Cairo, Cairo, Egypt*

- Developed front-end components for the internal Budget Module using Next.js, and built REST APIs with Node.js to support business logic
- Integrated Smartsheet for project tracking and workflow automation; participated in on-site workshops with Smartsheet representatives

- Designed and implemented Python Quart APIs for a Reinsurance Pricing Proof of Concept, including inflation-factor calculations, Excel uploads to Azure Data Lake Storage Gen2, and claims-triangle generation; collaborated with UK engineers through pair programming and code reviews
- Developed and deployed Python Quart APIs on Azure Functions to manage contract documents and metadata in Azure Data Lake Gen2 and Azure Cosmos DB, enabling uploading, soft-deleting, and metadata retrieval; supported Contract Wording Proof of Concept through Agile collaboration with UK engineers

Projects

FebriLogic | *FastAPI, Postgres, Streamlit* July 2025 – November 2025

- Developed JWT-based authentication APIs for user registration and login via email and password
- Designed and implemented patient management APIs, including adding new patients, updating patient information, inserting additional medical data (negative disease tests, symptoms, biomarkers), and retrieving associated patients
- Implemented APIs for retrieving and caching reference information including countries, diseases, symptom metadata, and biomarker metadata
- Deployed the FastAPI back-end, PostgreSQL database, and Streamlit front-end on Render for production availability, and additionally hosted the Streamlit front-end on Streamlit Community Cloud to increase redundancy and accessibility

Technical Skills

Languages: C++, Python, Bash, SQL (Postgres, MySQL), HTML/CSS, JS
Frameworks: Spring Boot, React, Tailwind, FastAPI, Next.js, Node.js, Angular
Developer Tools: Git, Docker, Google Cloud Platform (GCP), Microsoft Azure, VS Code, Visual Studio